RAJALAKSHMI ENGINEERING COLLEGE

Teja Shree U 251001282-2025-IT

T2

Started on	Monday, 31 August 2026, 12:53 PM
State	Finished
Completed on	Monday, 31 August 2026, 12:54 PM
Time taken	1 min 8 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Divide and Conquer: find the index of first 0
int findFirstZero(int arr[], int low, int high)
{
    if (low > high)
        return -1;

    int mid = low + (high - low) / 2;

    if (arr[mid] == 0)
    {
        // If this is the first element or previous element is 1
        if (mid == 0 || arr[mid - 1] == 1)
            return mid;

        return findFirstZero(arr, low, mid - 1);
    }

    return findFirstZero(arr, mid + 1, high);
}

int main()
{
    int m;
    scanf("%d", &m);

    int arr[m];

    for (int i = 0; i < m; i++)
        scanf("%d", &arr[i]);

    int firstZero = findFirstZero(arr, 0, m - 1);

    int zeroCount;

    if (firstZero == -1)
        zeroCount = 0;
    else
        zeroCount = m - firstZero;

    printf("%d\n", zeroCount);

    return 0;
}
```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

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Teja Shree U 251001282-2025-IT

T2

Started on	Monday, 31 August 2026, 12:58 PM
State	Finished
Completed on	Monday, 31 August 2026, 1:00 PM
Time taken	2 mins 46 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int majorityElement(int nums[], int left, int right)
{
    // Base case
    if (left == right)
        return nums[left];

    // Divide
    int mid = left + (right - left) / 2;

    int leftMajority = majorityElement(nums, left, mid);
    int rightMajority = majorityElement(nums, mid + 1, right);

    // If both halves have the same majority
    if (leftMajority == rightMajority)
        return leftMajority;

    // Count occurrences of leftMajority and rightMajority
    int leftCount = 0;
    int rightCount = 0;

    for (int i = left; i <= right; i++)
    {
        if (nums[i] == leftMajority)
            leftCount++;

        if (nums[i] == rightMajority)
            rightCount++;
    }

    // Combine
    if (leftCount > rightCount)
        return leftMajority;
    else
        return rightMajority;
}
```

```
}

int main()
{
    int n;
    scanf("%d", &n);

    int nums[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &nums[i]);

    int result = majorityElement(nums, 0, n - 1);

    printf("%d\n", result);

```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

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Teja Shree U 251001282-2025-IT

T2

Started on	Monday, 31 August 2026, 1:01 PM
State	Finished
Completed on	Monday, 31 August 2026, 1:11 PM
Time taken	10 mins 17 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int arr[], int low, int high, int x)
{
    // No element <= x
    if (low > high)
        return -1;

    int mid = low + (high - low) / 2;

    // If arr[mid] == x, it is the floor
    if (arr[mid] == x)
        return arr[mid];

    // If arr[mid] < x, it can be the floor
    if (arr[mid] < x)
    {
        int result = findFloor(arr, mid + 1, high, x);

        if (result == -1)
            return arr[mid];
        else
            return result;
    }

    // If arr[mid] > x, search left half
    return findFloor(arr, low, mid - 1, x);
}

int main()
{
    int n, x;

    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &x);

    int result = findFloor(arr, 0, n - 1, x);

    printf("%d\n", result);

    return 0;
}
```

	Input	Expected	Got	
✔	6 1 2 8 10 12 19 5	2	2	✔
✔	5 10 22 85 108 129 100	85	85	✔
✔	7 3 5 7 9 11 13 15 10	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

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Teja Shree U 251001282-2025-IT

T2

Started on	Monday, 31 August 2026, 1:11 PM
State	Finished
Completed on	Monday, 31 August 2026, 1:12 PM
Time taken	33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int binarySearch(int arr[], int low, int high, int key)
{
    if (low > high)
        return -1;

    int mid = low + (high - low) / 2;

    if (arr[mid] == key)
        return mid;

    if (arr[mid] < key)
        return binarySearch(arr, mid + 1, high, key);

    return binarySearch(arr, low, mid - 1, key);
}

int findPair(int arr[], int n, int x, int index)
{
    // Base case
    if (index >= n)
        return 0;

    int required = x - arr[index];

    // Search for the required element after current element
    int pos = binarySearch(arr, index + 1, n - 1, required);

    if (pos != -1)
    {
        printf("%d\n", arr[index]);
        printf("%d\n", arr[pos]);
        return 1;
    }

    // Recursively check next element
    return findPair(arr, n, x, index + 1);
}

int main()
{
    int n, x;

    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    scanf("%d", &x);
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course

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Teja Shree U 251001282-2025-IT

T2

Started on	Monday, 31 August 2026, 1:12 PM
State	Finished
Completed on	Monday, 31 August 2026, 1:13 PM
Time taken	34 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

For example:

Input	Result
5 67 34 12 98 78	12 34 67 78 98

Answer:

```c
#include <stdio.h>

void quickSort(int arr[], int low, int high)
{
    if (low < high)
    {
        int i = low;
        int j = high;
        int pivot = arr[low];

        while (i < j)
        {
            while (arr[i] <= pivot && i < high)
                i++;

            while (arr[j] > pivot)
                j--;

            if (i < j)
            {
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }

        // Place pivot in its correct position
        int temp = arr[low];
        arr[low] = arr[j];
        arr[j] = temp;

        // Recursively sort left and right parts
        quickSort(arr, low, j - 1);
        quickSort(arr, j + 1, high);
    }
}

int main()
{
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    quickSort(arr, 0, n - 1);

    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);

```

	Input	Expected	Got	
✔	5 67 34 12 98 78	12 34 67 78 98	12 34 67 78 98	✔
✔	10 1 56 78 90 32 56 11 10 90 114	1 10 11 32 56 56 78 90 90 114	1 10 11 32 56 56 78 90 90 114	✔
✔	12 9 8 7 6 5 4 3 2 1 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	1 2 3 4 5 6 7 8 9 10 11 90	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course